

CONVERSATIONAL CLUSTERING

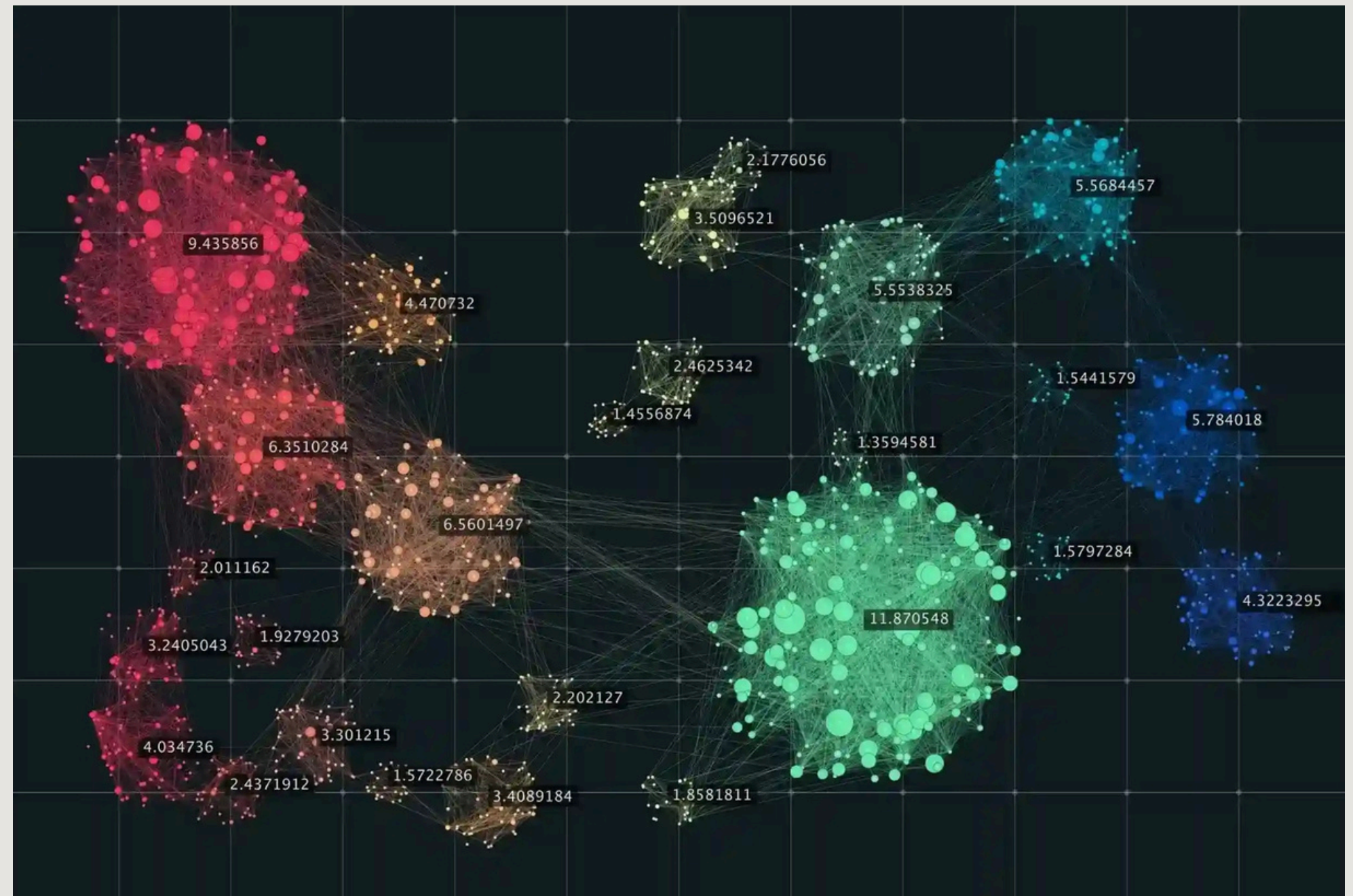
PROBLEM STATEMENT

TECHNICAL KNOWLEDGE

Users often lack technical knowledge about clustering techniques

FINETUNING

Translating high-level intents into technical choices places the cognitive burden in the wrong place



RESEARCH QUESTIONS

QUESTION 1

What properties of the conversation affect how the user's intent is captured?

Specifically: which question type is the best to ask, in terms of conversation length (number of turns) and total tokens exchanged?

QUESTION 2

How faithful is the proposed result, given the preferences of the user?

If the user requires additional changes (i.e. merging or splitting clusters), can the system update its representation to meet the user's preference?

OTHER QUESTIONS

QUESTIONS THAT WON'T BE ADDRESSED

- Will the system be able to process large datasets (e.g. hundreds of thousands or more datapoints)?
- Can/does the system work on non-text data (tabular, image, time-series)? The current design assumes text input throughout
- How does the system behave when the user has no strong clustering preference and cannot answer the LLM's questions meaningfully?
- How robust is the system to vague, ambiguous, or deliberately unhelpful user answers?
- Can the system update the clustering as new data arrives, without restarting the conversation?
- Can the system explain why a specific datapoint was assigned to a specific cluster in terms the user understands?

RELATED WORK: CONVERSATION

RELEVANT PAPERS

- Eliciting Human Preferences with Language Models
- TO-GATE: Clarifying Questions and Summarizing Responses with Trajectory Optimization for Eliciting Human Preference
- **Asking Clarifying Questions for Preference Elicitation With Large Language Models**

RELATED WORK: EMBEDDINGS

RELEVANT PAPERS

- Improving Text Embeddings with Large Language Models
- NV-Embed: Improved Techniques for Training LLMs as Generalist Embedding Models

RELATED WORK: CLUSTERING

RELEVANT PAPERS

- Large Language Models Enable Few-Shot Clustering
- Dial-In LLM: Human-Aligned LLM-in-the-loop Intent Clustering for Customer Service Dialogues
- Optimized Algorithms for Text Clustering with LLM-Generated Constraints
- Summaries as Centroids for Interpretable and Scalable Text Clustering

ARCHITECTURE

1. CONVERSATION

The users explains its clustering preferences to the LLM. If needed, clarifying questions are asked by the model, to better frame the user's intentions

2. EMBEDDINGS GENERATION

Use NV-Embed to encode all the documents in the dataset. The user's preferences are also taken into account.

3. CLUSTERING

A clustering algorithm is run on the embedded documents. Strategies such as summary-as-centroids or keyphrase generation can be used to improve the results.

4. CLUSTER REFINEMENT

The users asks the LLM to merge or split clusters based on some conditions, to achieve a solution that better suits its needs

DATASETS

STACKOVERFLOW

Around 20k short texts (question titles).
Clustering can be based on programming language, architectural vs debugging questions, etc.

GOEMOTIONS

Around 58k Reddit comments.
Clustering can be based on the emotion of the comments.

EVALUATION

INTERVIEW QUESTIONS TO EVALUATE

- yes/no questions
- multiple choice questions
- open-ended

CONVERSATION QUALITY

- **efficiency:** turn to convergence and tokens exchanged
- **question clarity:** number of tokens used by the user to generate an answer
- **faithfulness:** does the final result reflect the user preferences?
Assessed using an LLM-as-judge, rating the clustering assignments on a Likert scale (from 1 to 5)